Domain Research Note

Project Title

ScholAR: Local LLM Research Paper Assistant

Domain

Indic NLP and Agentic AI, applied to academic paper understanding and research assistance.

Real Problem

Research papers are the main source of new progress in AI and machine learning, but they are difficult to read for students and early researchers. A single paper contains the problem statement, related work, method, experiments, results, limitations, and implementation details. The challenge is not only reading the PDF, but understanding what matters, how sections connect, and whether the paper is relevant for a project.

This problem is stronger in fast-moving areas like Generative AI and LLMs, where many new papers appear on arXiv every day. Manual reading is slow, while copying parts of a PDF into a chatbot is unstructured and often loses context.

Who It Affects

This affects undergraduate and postgraduate students, research interns, early machine learning engineers, and GenAI project teams. These users may understand basic AI concepts, but often struggle with dense research writing, mathematical explanations, and long experimental sections. In the Indian academic context, many students also have limited access to expensive cloud APIs, so a local-first tool is more practical, private, and affordable.

Why Existing Solutions Fall Short

Normal PDF readers can display and search papers, but they do not explain them. General chatbots can explain selected text, but the user has to manually copy sections and manage context. Basic chat-with-PDF tools use retrieval over chunks, which works for simple questions but can miss answers that require connecting the abstract, method, results, and limitations together.

Many existing tools also depend on cloud models, which creates cost and privacy concerns. Most of them answer questions reactively instead of guiding the user through a structured study process with study goals, citations, and verification.

Proposed Approach

ScholAR is a local-first research paper assistant inspired by PaperBreakdown. In Milestone 1, users can search arXiv papers, view paper cards, open a paper detail modal, download the selected PDF, extract page-wise text using PyMuPDF, create chunks, and study the paper using a local Ollama Qwen model (Qwen 3.5: 9B).

The system uses a split workspace with the PDF on one side and an AI study panel on the other. The backend retrieves paper metadata, processes the PDF, stores pages and chunks locally, retrieves relevant chunks for a user question, and generates grounded answers with page-level citations. It also generates study goals so users can understand the paper step by step.

Why This Approach Is Justified

This approach is justified because the main problem is structured paper understanding, not just text generation. A local retrieval plus LLM pipeline is simple, explainable, and suitable for student hardware. It also avoids sending papers to cloud APIs.

The project is inspired by Recursive Language Models, where long documents are treated as external information instead of being fully placed inside the model context. ScholAR applies this idea in a safe Milestone 1 form by storing the paper externally, retrieving useful chunks, and asking the local model to reason over selected parts.

For Milestone 1, the aim is to prove the core pipeline works: arXiv search, PDF download, text extraction, chunking, retrieval, study goal generation, and grounded local LLM question answering. This creates a strong base for the final version, where better retrieval, RLM-style multi-step analysis, knowledge checks, and quantitative comparison can be added.